



The Ephemeris



December 2017

Volume 28 Number 04 - The Official Publication of the San Jose Astronomical Association

Dec 2017 - Mar 2018 Events

Board & General Meetings

Saturday 12/2, 1/6, 2/3, 3/3, 3/31

Board Meetings: 6 -7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 12/3, 1/7, 2/11, 3/4

Solar Observing (locations differ)

Sunday 12/3, 1/7, 2/11, 3/4

Intro to the Night Sky Class

Houge Park 1Q In-Town Star Party

Friday 12/22, 1/26, 2/23, 3/23

Houge Park 3Q In-Town Star Party

Friday 12/8, 1/12, 2/9, 3/9

RCDO Starry Nights Star Party

Saturday 12/9, 1/13, 2/10, 3/10

Imaging SIG Mtg

Tuesday 12/19, 1/16, 2/20, 3/20

Astro Imaging Clinics

(at Coyote Valley OSP)

Saturday 12/16, 1/13, 2/10,

Coder's Meeting

Sunday 1/21, 2/18, 3/18

Quick STARt (by appointment)

Friday 12/9

Binocular Star Gazing

No Winter Events

Unless noted above, please refer to the SJAA Web page for specific event times, locations and possible cancelation due to weather.

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Cover Photo: August 21, 2017 Eclipse

Location: Mitchell, Oregon

By SJAA Members: **Bill & Susan O'Neil**



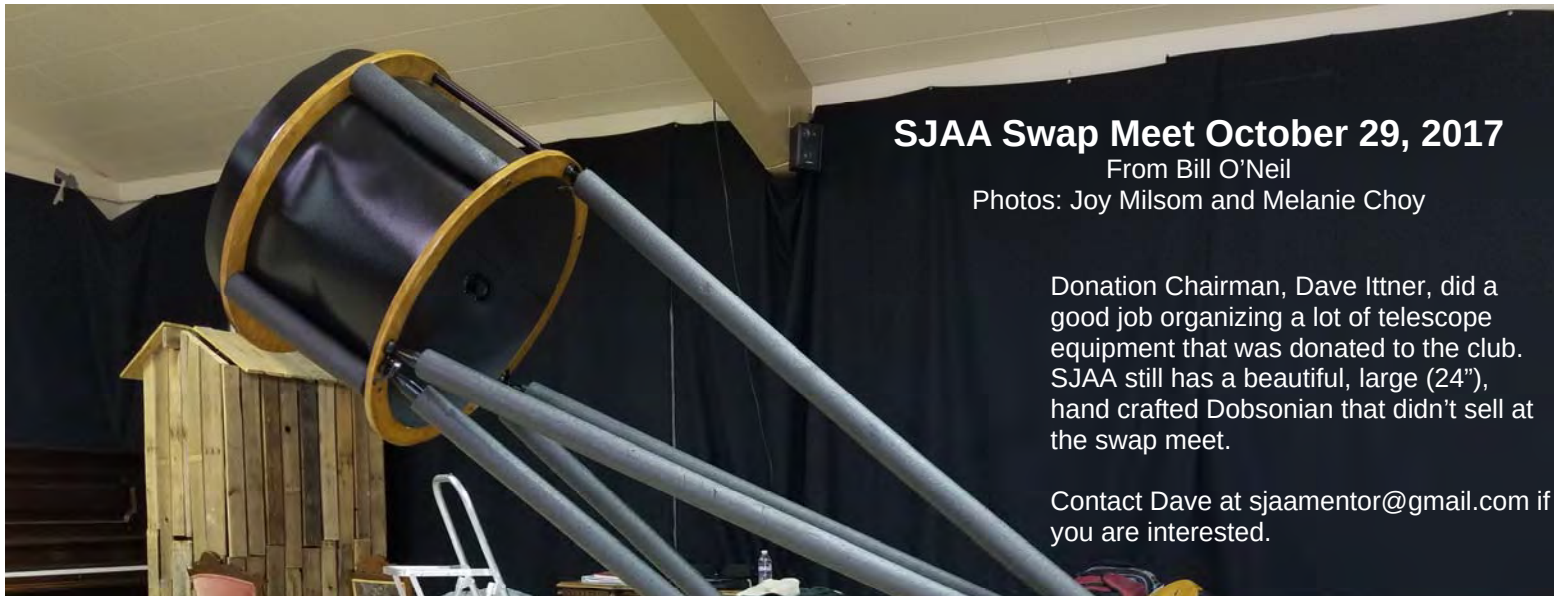
SJAA Swap Meet October 29, 2017

From Bill O'Neil

Photos: Joy Milsom and Melanie Choy

Donation Chairman, Dave Ittner, did a good job organizing a lot of telescope equipment that was donated to the club. SJAA still has a beautiful, large (24"), hand crafted Dobsonian that didn't sell at the swap meet.

Contact Dave at sjaamantor@gmail.com if you are interested.



Jim Van Nuland checking out the merchandise



Above: Marianne and Terry
Below: Bill O'Neil, Joe Lin, David and Vini



Doors Open!

SJAA held its annual Fall Swap Meet on Sunday, Oct. 29, 2017 at the Houge Park Hall. There was a large queue of buyers waiting for the doors to open at noon on Sunday.

The SJAA Swap Meets are always a good opportunity to reconnect with friends from SJAA and other Bay area clubs, like Eastbay Astronomical Society, Tri-Valley Stargazers and the Santa Cruz Astronomy Club. Thank you all for making the trip to San Jose and making our Swap Meet a success. SJAA brought in nearly \$2,000 (and over half of that was from the sale of donated equipment).



Stargazing party at the Pinnacles, a huge hit

posted on Wed, 09/20/2017 - 02:28pm by Becky Bonner-Leland, Reporting for BenitoLink



*Through the telescopes one could see the Lagoon and Trifid Nebulas.
Photo courtesy of Ed Wong.*



*The Andromeda Galaxy was visible for participants to see.
Photo courtesy of Ed Wong.*



*The event Saturday night took place behind the visitor center on the East Side of the park.
Photo credit Becky Bonner-Leland*

Editor's Note: SJAA's Ed Wong invited Becky Bonner to attend the 9/16/2017 NPS Pinnacles Star Party where she was able to witness first hand the wonders of the night sky at Pinnacles National Park. The below article was published a few days later on the BenitoLink website: <https://benitolink.com/>

Sept.16, 2017 roughly 200 people came out to the Pinnacles National Park to participate in their Stargazing Party. While there, participants were able to view different sights, such as Saturn, through ten telescopes available.

Though the Pinnacles National Park has been a long time favorite destination and source of natural beauty for San Benito County, its Stargazing Parties are a new addition. Started on the West Side of the park in 2015, the Pinnacles decided to bring it to the East Side to increase participation. Astronomer docent Ed Wong explained, "The East Side has a lot more people. Originally, we did the West Side trying to get more people to come out. The West Side has no camping so it's less busy. On the West Side, we get 50 people at best. On the East Side, we had 200."

As the city of Hollister grows in population, and open spaces continue to be filled with new homes and street lights, opportunities to reconnect to nature and find an open spot to look up and wonder without light pollution decreases. When talking to Wong about why the Pinnacles decided to incorporate Stargazing Parties at the park he stated, "The thing is I kind of pursued them to start something, but then, because the Pinnacles is a National Park and there is an awareness of conservation, the dark sky is another area of conservation they want to promote. Light pollution is quickly taking over everything and it effects the animals and wildlife. Preserving dark skies is something that can happen overnight if people want... It makes it special because most people nowadays don't see the night sky because it is so polluted." This reason did not fall on deaf ears as roughly 200 people attended the event Saturday night, all without being able to use cameras or cellphones to capture images to preserve the integrity of the experience.

People were free to walk around to different telescopes set up in the open area behind the Visitor's Center at the park. While going to all ten of the telescopes available, each with a different view, one was able to see Saturn, Neptune, binary stars, supernova remnants, the Lagoon Nebula, and other star clusters. The night sky was so clear at the site that one could easily see the Milky Way and Big Dipper with the naked eye. However, one of the best features of the night was talking to the ten astronomers on site who came out to help put on the Stargazing Party. Starting at 8:30 p.m., the event lasted an hour and a half. Each astronomer showed passion and knowledge as they not only described what one was seeing, but also answered any other related questions that were asked.

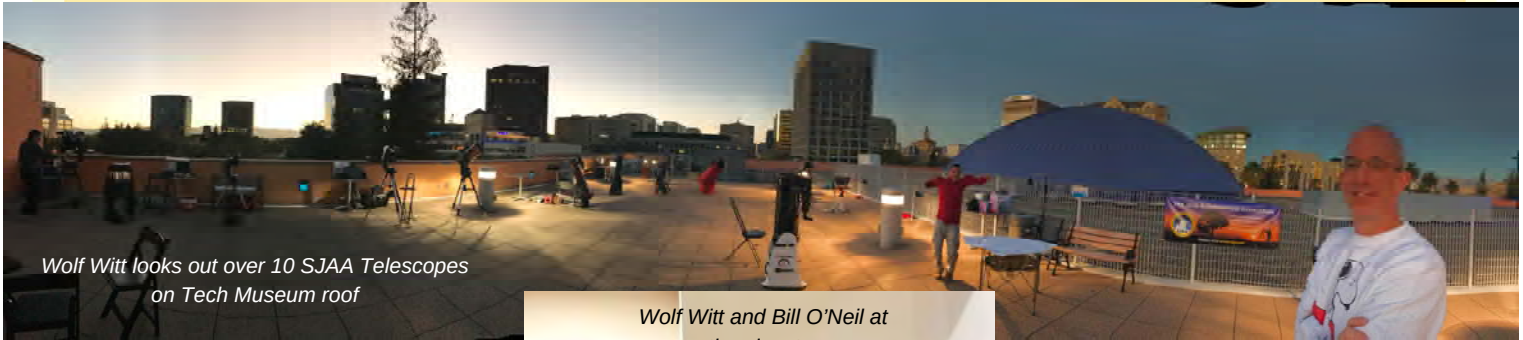
When talking to participant Kelli Simmons of Gilroy about why she decided to come out to the event she stated, "I am thrilled I accepted a friend's invitation to go stargazing. It is worth the drive to get out of the city lights. I am overwhelmed by the beauty of the star filled night sky. The high power telescopes are amazing and the astronomers are very knowledgeable and friendly." Ms. Simmons was not the only person who traveled from outside of the area to try to catch a glimpse of the Lagoon Nebula. Though no official data was taken on who came out Saturday night, astronomer Ed Wong shared, "Since the event is open to the public and there is no forced sign-ins we do not have documented data. However, we do ask people at random to get ideas of where people come from. From last night, I know there we people came from the following areas: Mountain View, Palo Alto, San Jose, Los Gatos, Hollister, Gilroy, Morgan Hill so we do have quite a reach for the program."

Although this will be the last Stargazing Party for the season at the Pinnacles on the East Side of the park, there will be another event on the West Side of the park October 14th put on by a similar group of astronomers. Wong and fellow astronomers find joy in sharing their passion for the night sky with others and work hard as a collective team to bring these events to fruition. "To share with people I get gratification too. Hopefully, I can help other people realize the world is bigger than what they can experience in a regular day," Wong explained. While looking through the astronomers' telescopes with no obstruction from light pollution by nearby cities, one could not help but agree with Wong in not only the brilliance of the night sky, but being reminded how vast our world and universe really are.



San Jose Tech Museum Star Party

From Bill O'Neil



Wolf Witt looks out over 10 SJAA Telescopes on Tech Museum roof

SJAA Hosted a Night Star Party on the San Jose Tech Museum Roof

The San Jose Tech Museum of Innovation sponsored a huge Convention for the Directors and Staff of many Science & Technology Museums throughout the USA and World. Nearly 2,000 Association of Science Technology Centers (ASTC) visitors came to the Tech Museum and the San Jose Convention Center, to discuss ways to draw the public to visit Science Museums to encourage curiosity, experience a hands-on science activity, and open a window into lifelong learning (and teaching).

As part of the big Opening Night Celebration, the Tech Museum invited the San Jose Astronomical Assoc. (SJAA), to host an evening Star Party up on the Tech Museum Roof on Saturday night, Oct 21, from 7:30pm to 11:30pm. SJAA brought 10 large telescopes (with knowledgeable astronomers) up on the Roof to show the night sky to the ASTC guests.

SJAA put out a Call for Telescopes on its Yahoo Observers Group website. Our Astronomers included: Wolf Witt, Rashi Girdhar, Rob Chapman, Gary Mitchell and Bill & Susan O'Neil. Because we expected 1,000 to 2,000 visitors, Bill also reached out to the Eastbay Astronomical Society and they sent 4 telescopes and operators including: Nate McKenna, Susan Rambo, Jim Gibbs and their Outreach Coordinator, Ray Howard.



Wolf Witt and Bill O'Neil at the elevator

The crowd was treated to clear views of Saturn and it's Rings until it went behind low buildings after 9pm. We enjoyed several Globular Clusters, M2, M11, and M13: (hundreds of thousands of stars, gravitationally bound), plus our nearest Galaxy neighbor, M31, the Andromeda Galaxy, 'only' 2.5 million Light Years away. It was a Great Nighttime Event!

Our Outreach astronomers had a ball. It

was a little bit like Preaching to the Choir, all the attendees were seriously into science and knew a lot about astronomy, but it had been a while since they actually looked into a telescope. After Saturn set, Bill set up his small telescope to look at the Seven Sisters, the Pleiades Open Cluster (Messier object: M45, see photo attached). It is a tiny Little Dipper (Not the one with Polaris in the Handle, but still very beautiful). It is actually a relatively Large, naked eye

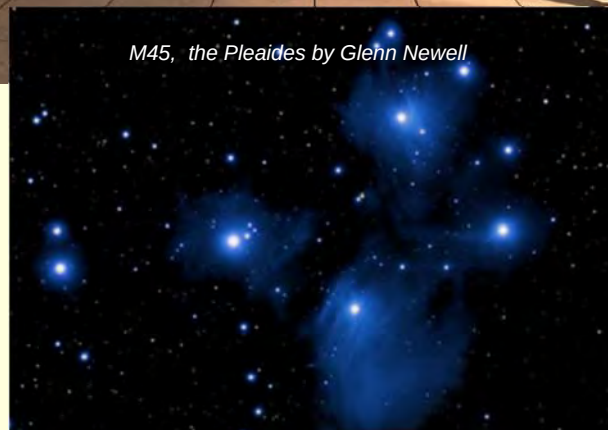
object in the night sky, (2°, almost 4 full Moons wide) so the Big telescopes can't see all of it in the eyepiece. But Bill's 5" scope and a focal reducer with a 40mm eyepiece, fit it all in.

Bill was having fun by showing them the Pleiades and asking people what kind of car they drive? He

would get a weird look, but would ask if they ever drove a Subaru? Then he would explain that the word for the Pleiades star cluster in Japanese is "Subaru" and the Subaru Logo is a stylized 6-star version of M45. Very Cool!!! Something for the Geeks to talk about at the Bar!



SJAA astronomers by the club banner



M45, the Pleiades by Glehn Newell



Of Two Minutes and Two Seconds

Poem written by Monica Korde
Photos by Nikhil Tungare

*Lunar tales
Solar flares
A celestial journey unfolds.*

*Bright light
Out of sight
Farewell, dear Sun.*

*Fictional skies
Hungry eyes
Shadows race across.
Ring ceremony
Perfect harmony
Behold the astral engagement.*

*Cosmic coronation
Lost in transformation
Time stands transfixed.*

*Reverse, Retrace
It's a Final embrace
The astronomical drama ends.
Unveil, Unwind
Nothing's left behind
Except -*

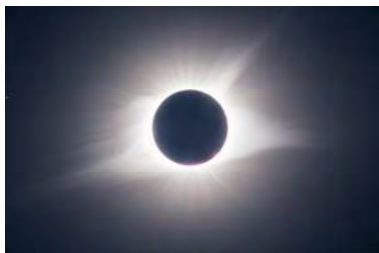
An ephemeral memory and our Eclipsean T-shirts.



SJAA Member Observation Post + Quick Start

Total Solar Eclipse 2017 at the Oregon Star Party

From Gary Chock



I attended the 2017 Oregon Star Party (8/17 to 8/22), including the Total Solar Eclipse (TSE) on Monday 8/21/2017. SJAA members Donald Gardner, Allen Frohardt, and Mark Striebeck were also there. Patchy clouds and smoke threatened in the days leading up to the TSE, with Sunday night starting out as cloudy. Thankfully, it cleared after midnight and we enjoyed the TSE with only just a bit of high cirrus.

I wholeheartedly agree with all the comments about totality being above and beyond any partial eclipse: WOW! Amazing! Wondrous...

We noted the temperature drop before and temperature rise after totality. A slight breeze from the west with the cooling and a reciprocal slight breeze from the east with the heating. Measurements noted an 8 degree drop from the temperature at the start. Based on the morning temperature rise curve interpolated from before 1st Contact (C1) and after 4th Contact (C4), they estimated a -15 degree difference at the time of totality.

Birds started chirping as it dimmed. Folks noted that they stopped 5 minutes before 2nd Contact (C2). They restarted chirping as it brightened about 15 minutes after 3rd Contact (C3).

We had no reports of shadow bands, probably due to the high cirrus. However, that high thin deck allowed for noticing the westward-approaching and eastward-departing darkness.

I visually noted the Diamond Ring and Baily's Beads effects at C2 and C3. I took time to take in the 360 degree horizon twilight at mid-totality. I identified Venus and Denebola, but I was so mesmerized by the corona and prominences that I missed the other planets and stars.

The corona streamers were asymmetrical as expected near the solar minimum. I had extended views of the corona and prominences naked-eye and with my 10x42 binoculars. I held on to the binocular views of Baily's Beads and Diamond Ring effect, lingering as totality ended...

TSE 2024 Totality or Bust!

Quick Start Accessory Cases Upgraded

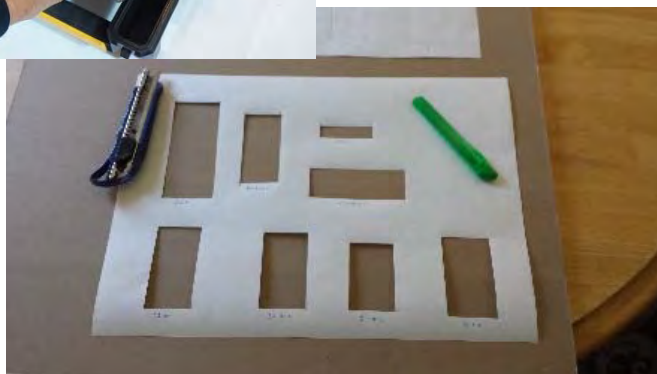
Posted by Dave Ittner August 25, 2017

The old Quick Start accessory cases were created from Harbor Freight Tool cases that had pluck foam in them. Over the years this foam had been tugged on and became loose leaving much larger openings than what was originally created for items like eyepieces. This allowed these items to move around far too much. It was time for new inserts.

I made a post in our Observers Yahoo group and got a lot of good ideas. The one that made the most sense to me was Kaizen Foam from Fastcap.

www.fastcap.com/estore/pc/Kaizen-Foam-p13435.htm

What makes this foam special is that it is constructed of a number of thinner sheets....such that after you make your cut, all you have to do is peel off sheets to the depth you want. I decided to make a template since I had 8 cases to make (see below).



I cut out each area, overlaid the template onto the foam, and marked the edges where I needed to cut. After cutting out each area I then peeled the foam down to an acceptable depth.

What prompted all of this was we decided to purchase 20mm Plossl eyepieces to add to each kit. Now there is a 32mm, 25mm, 20mm, 8mm eyepiece. As well as a 2X barlow, a moon filter, a red flashlight, and a Rigel Quick Finder. These cases will now last for a very long time.



O'Neil Total Solar Eclipse Adventure

From Bill O'Neil

Hope you all got to see a least a Partial Solar Eclipse last Aug. 21, 2017. Here is our story of the O'Neil Family Total Solar Eclipse Adventure:

We began looking for Hotels and Airbnb Rentals about a year before the event. Many of the SJAA & SCAC crowd were going to be in Madras, OR, but they were expecting up to 200,000 visitors at that site. Fortunately, friends from Eastbay Astronomical Society (EAS) had found the Painted Hills Vacation Resort in Mitchell, OR (population 121, about 50 miles due East of Madras). In addition to several large cottages, the owner had also converted her Art Studio into a small one-room cabin, which was just perfect for Bill & Susan O'Neil and their daughter, Shannon, & husband, Stephen, who drove down from Seattle, WA. SJAA Director, Wolf Witt, also planned to join us, but decided to rent a nearby Airbnb for him and his guests.

We drove up to Oregon early (and stayed with relatives in Yreka) then drove up to Mitchell before the crowds. FYI, near Crater Lake, in Klamath Falls, OR there is a fantastic Native American Indian Museum (the "Favell Museum" is at 125 W. Main St, open 10am-4pm, Tues – Sat.) and features an extraordinary collection of thousands of arrow-heads, obsidian knives, spear points, primitive ancient stone tools, native clothing, intricate beadwork, basketry, pottery and more. Truly worth every minute of your visit (allow 2hr +). In Mitchell, we also visited the nearby Painted Hills (very colorful hills of red, green and black) and the John Day Fossil Beds National Monument, both fun adventures.



Shannon and Stephen—Glasses ON Mitchell, Oregon



O'Neil Clan - Mitchell, Oregon



O'Neil Cottage - Mitchell, Oregon

The "O'Neil Cottage" was high on a hill over-looking the town of Mitchell. Outside was a large open field (at 3,000 ft.) that had a perfect view of the south-east rising Sun (and solar eclipse). We and EAS set up our telescopes the night before the eclipse and hosted an evening Star Party for the residents of Mitchell. We got up early Monday for a final check out of our telescopes and photography equipment.

'O'Neil Adventure' Continued on following page

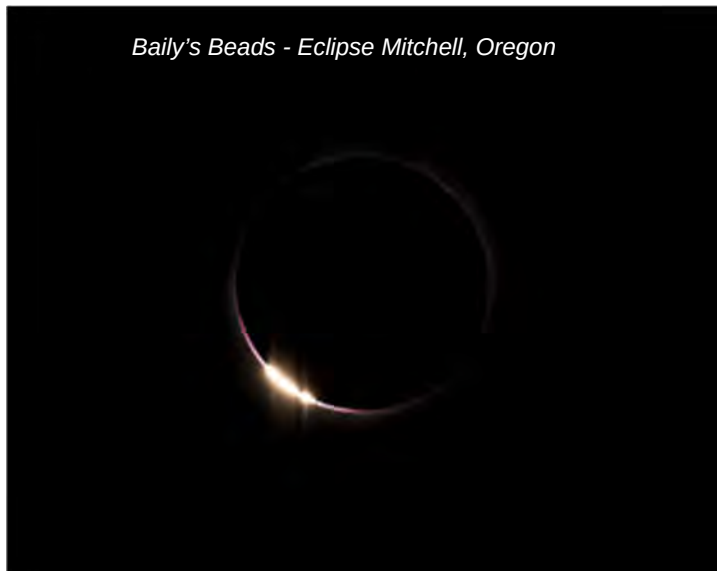


Diamond Ring - Eclipse Mitchell, Oregon



O'Neil Total Solar Eclipse Adventure—continued

Baily's Beads - Eclipse Mitchell, Oregon



Totality - Eclipse Mitchell, Oregon



See front cover for an enlarged view of the solar flares.

Corona - Eclipse Mitchell, Oregon



End Baily's Beads - Eclipse Mitchell, Oregon



Susan was using her Nikon D-810 full-frame DSLR with a 80mm-400mm telephoto lens mounted on a Celestron NexStar GPS Tracking Mount that Wolf loaned us. (Wolf had his Lunt Hydrogen-alpha telescope AND a small refractor scope, with an iOptron mount, to watch the Eclipse).



End of Eclipse - Mitchell, Oregon

Susan & Bill had also downloaded "Solar Eclipse Maestro" software by Xavier Jubier who helped Google & NASA with their interactive 2017 Eclipse Maps. This software is for Macs and takes a GPS data input and works with your Script of the bracketing shots you want to take at the key points in the Eclipse. The Camera is tethered to the Computer and is "hands-free" once the Eclipse begins.

We received a lot of help from Rob Hawley, Gert Gottschalk, Swaroop Shere, and Wolf Witt, setting up and debugging the Eclipse Maestro app. We were very pleased with the photo results. See the Diamond Ring, Baily's Beads, Totality, Flares, Corona and End of Eclipse photos, presented here-in.



Eclipse Toast - CORONA Beer

And how do you celebrate a successful Total Solar Eclipse? With a Toast of "Corona" beer, of course!

See you in Texas in 2024?



SJAA Member Observation Posts

Calstar Reports - Dave Ittner Message 1 of 3, Sep 24, 2017

Hey, for those who couldn't make it to CalStar this year, we'd love to hear your thoughts about the experience, so please take a moment to share.

Prabath Gunawardane Message 2 of 3, Sep 29, 2017

A bit late but getting around to jotting down my thoughts. I haven't been to either CalStar or GSSP before so this was my first big multi-day star party (not counting a time we did the SJAA group camping at Pinnacles).

I was there from Wednesday afternoon to Sunday morning and had a wonderful and memorable time. I've only seen darker skies when I've been backpacking in the Sierras but this was easily the darkest skies I've been able to setup my telescope and observe under. The Milky Way was very bright and even extended to a fuzziness in Gemini.

The first night I spent a couple of hours re-familiarizing myself with all the constellations in the fall sky, even the less conspicuous ones. Then realized that the battery pack for my 10" GoTo Dob had a loose connection. Still it gave me a good chance to practice star hopping to pick-up some nice globular clusters in Ophiuchus and Sagittarius. I also made good use of my binos (10x30 Canon IS) for some wonderful wide field views. The seeing was great until about 1:30am when some clouds started rolling in from the northeast. The second night started off well, with clear skies and slightly colder than the first. The seeing was good and I went on a Messier binge since I was able to jury rig the battery pack for the GoTo again. I was able to get some really nice views of Lagoon, Trifid, Swan and Eagle nebulas. Just past midnight though it started getting really cold. I was woefully under layered and decided to just call it a night around 1am.

Third night was the best for me, layered up and observing until 4:30am. Spent a lot of time looking at the faint stars inside the Orion nebula and exchanging some nice views with the neighbors. I starhopped my way through the fall messier list and used the GoTo for dozens more. Continued working my way

through that list for the 4th night, but the highlight of the evening was the Atlas V launch as others have also mentioned. I was able to follow it through my binos and could clearly see the jet plumes, the contrail patterns and the separation. It was definitely an experience. As a bonus, I tracked down Uranus and Neptune using the charts.

All in all it was an amazing four days; great observing, wonderful and friendly community and generally very cooperative weather. It wasn't too hot during the daytimes, and apart from the second night... not too cold at night. All the people I met were super friendly and I had a very enjoyable time socializing during the day or sharing views during the evening. Also it was extremely convenient to camp right next to the scope and have the flexibility to stay up until whatever time you wanted to without having to worry about the drive back or the spectre of a workday coming up. I think this has made me a firm believer in multi-day star parties under dark skies and definitely hope to return to CalStar next year.

Stephanie Drews - Message 3 of 3, Sep 30, 2017

A little late as well, but here goes. I had a chance to attend CalStar for the first time this year with my husband and 2-year-old daughter. I was very grateful that my husband agreed to make it a family trip since I rarely get to get out and observe these days due to child and work obligations. In fact I don't think I've had a chance to come to any SJAA club meetings or meetups lately; I'm probably a bit of a stranger to you guys. So this was a big deal for me. It was the first group observing chance I'd had since attending Golden State Star Party several years ago (pre-baby), and only one of a handful of times I've gotten a chance to observe in the state of California, after moving here from Oregon 7 years ago.

It was also the first light for my 16" Meade Lightbridge Dob. Some nice folks helped me with collimation since my Orion collimator turned out to be useless and I got to enjoy a wonderfully upgraded viewing experience over my previous 10" Dob. I was very pleased to

have all 4 nights be clear and cloudless - a wonderful experience after many years of observing from the Pacific Northwest's skies! To be honest, I thought the transparency, darkness and clarity left just a tad to be desired. But again, it seems to be a tradeoff for a better chance of clear skies down here in CA, vs Oregon.

I am working on my Herschell 400 list, and started the trip with 112 objects remaining. Most of the objects left on my list are in the Winter/Spring skies, so I found it most productive to wake up around 2am and spend the early morning with Cetus, Monoceros, Lynx, Camelopardalis, Orion, Sculptor and a bit of Gemini and Cancer. NGC 908 in Cetus was a memorable hunt with a worthy reward at the end. Lynx gave me the most beautiful galaxies I've seen in a while-- NGC 2683 really stood out. NGC 2782 really gave me a chase but I found it. As it also happens, Lynx has several lovely double stars on the AL's Double Star program that I was easily able to split in my new giant telescope. Overall I was greatly pleased with the viewing experience. I revisited many old favorites to see what they looked at in my new scope and also knocked a dozen or so new items off my Herschell list.

I also thought the site itself was great. The weather was not too hot, stayed mostly in the low 80s or high 70s with some wind gusts and didn't get too cold at night. By some miracle the observing site had an amazing playground, perfect for my 2-year-old. She spent most of the time running around that thing. We were also able to take a day trip to Morro Bay on the third day. The hosts were also awesome, and cooked a complimentary breakfast buffet every morning. Everyone was very friendly and helpful-- the only slight drawback was the tight space conditions of the observing field. I've never seen scopes packed so closely together! Luckily everyone was very polite and accommodating and community oriented so it wasn't a problem, just had to be extra mindful of neighbors. I can see why folks are excited to move this event back to Lake San Antonio where it's much more roomy, but selfishly I hope it's at Whisper Canyon again just for the awesome playground :)

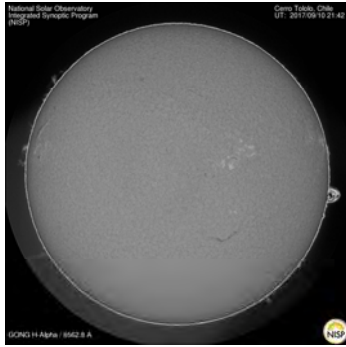


SJAA Member Observation Posts

ENORMOUS prominence on the sun right now

Mperry94 - Message 1 of 6 , Sep 10 4:11 PM

If you have a H-alpha telescope, take a look at the sun. It has an ENORMOUS prominence, quite unlike anything I have ever seen. Below is the Mauna Loa photo showing prominence at 3:30. I thought there must be something wrong with their camera, but my PST shows the same thing.



Rob Jaworski - Sep 10 4:14 PM

Bill O'Neil, Marianne and a few others are doing a solar thing this afternoon for the city parks dept. They are expecting a thousand people at the overall event. I sure hope they are showing off that splendid arch. That sure is fantastic!

Mperry94 - Sunday, September 10, 2017 4:20 PM

I just read on spaceweather.com for Sept 10;
"MAJOR SOLAR FLARE AND RADIATION STORM: Departing sunspot AR2673 erupted again on Sept.10th (1606 UT), producing a major X8-class solar flare. Protons accelerated toward Earth by the explosion are swarming around our planet now, causing a moderately strong solar radiation storm.

"A pulse of ultraviolet and X-radiation from the flare ionized the top of Earth's atmosphere, triggering a shortwave radio blackout over the Americas (link to map). That blackout is now subsiding. However, an even stronger blackout of shortwave radio communications around the poles is still underway (link to another map). Stay tuned for updates about this significant event."

Anand Rajagopalan - Sep 10 4:28 PM

This is awesome. Thanks for sharing! I have an H-Alpha that I have never had a chance to learn how to set up. I bought a Starfire 6" scope and a H-alpha filter for a nice deal but never got an opportunity to try it out yet. I will take any help in learning to set them up.

Sean McCauliff - Sep 10 5:02 PM

Thanks for pointing this out; truly an amazing prominence.

Marianne Damon - Message 6 of 6 , Sep 10 5:27 PM

Wow. I decided to take a look at the sun and put the solar filters securely on my 10x50 binos before deciding if I wanted to take out my bigger equipment. I looked at 4:43 pm and in just a few seconds I saw white flashes. I moved off the sun and double-checked my filters. Saw two more episodes of these flashes up until 4:45 when my instinct told me not to look anymore. I ran into the house and pulled up Space-weather and saw the below info. Then went to e-mail to write to this group, and saw all of these posts. I know I'm not Superwoman and do not have x-ray vision, but I definitely saw flashes of visible light through my solar filters! And, yes, AR2673, just around the visible limb, is where I saw the flashes.

Program Report for West Pinnacles Star Party 10/14/2017

From Ed Wong

To: Pinnacles East and West Star Party Volunteers

All, Thanks again for your help last night and throughout this year. I thought I'd share with you some feedback I got back from the Interp Ranger, Beth, on my program report to her this morning. If you haven't replied back to the volunteer appreciation BBQ please do. We worked hard for it.

----- Forwarded message -----

Elizabeth Hudick (Interp Ranger, Pinnacles National Park)

Date: Sun, Oct 15, 2017 at 2:32 PM

Re: Program Report for West Pinnacles Star Party 10/14/2017.

To: Ed Wong

Thanks for the feedback, Ed! I'm really glad to hear that everything went well. I appreciate all of your hard work this year, and I'm looking forward to seeing you at the Volunteer Appreciation BBQ on the 28th! Have a great day.

Beth Hudick
Park Ranger
Pinnacles National Park
5000 Highway 146
Paicines, CA 95043

Response: Gary Chock

Ed, Thought I would give you a recap of our activities after you departed....Wolf Witt and Tom Piller hung out with me as I packed up my gear. They left after 1am-ish. A little later, I hung out with Mark McCarthy who had set up his 20" Dob over behind the visitor's center. Besides looking though it every now and then, I was enjoying just naked-eye and binocular views. Conditions were mushy - Mark could only make out 4 of the Trapezium stars in M42 in his 20", so not too good. The Beehive Cluster was barely visible naked-eye. But on the positive side, the temperature was reasonable and the area back behind the visitor center building is nicely sheltered.

We were waiting for the crescent moon to rise, hoping to see Regulus come out from behind the dark side of the moon. Unfortunately, we discovered the horizon was probably too high to observe the event since the occultation was occurring at 3 degrees altitude, which was below our horizon. Around 3am, I ended up walking the new trail behind the visitor center to get around the hillside [and to a lower horizon] to try and see Regulus. From this new spot, I was able to catch the crescent moon (darkside, the "horns") rise above the tree line. Mark was there to observe Regulus's 12th mag white dwarf companion, a possibility due to the occultation, though his 20", but couldn't see it because it was quite low and blocked by the hill side. So this was a disappointment. After I saw Regulus and the crescent moon rise, I took a short nap, left at 4:10 am and got back to SJ by 5:45 am.



SJAA Member Observation Posts



Solar eclipse from 8000ft @ Eldorado National Forest

Tue, Aug 22, 2017 at 9:56 AM, Nikola Nikolov

I made it a bit beyond 80% eclipse and I found a new small spot for SJAA events.

Initially, I booked hotel at Oregon, but wildfires and traffic jams scared me, so I headed south. My plan B was to camp around Shasta, and then go to Crater Lake for the eclipse. Nah. Yreka was in smoke as well. At the end, I ended up close to Kirkwood ski resort off of highway 88.

I found a nice parking lot at an elevation around 8000 ft, where I set up. The location of the parking is very hidden. It is 10-12 miles from Kirkwood along highway 88. The location can fit about 10 cars on the parking. 360 degrees views everywhere. Light pollution was not a problem at all at this site.

I'm still planning on visiting Crater Lake; just waiting for wildfires to be fully contained and smoke to scatter.

August 22, 2017, 8:53 PM - Rob Jaworski

Scouting out new places is always cool. Do you have the coordinates of that place along highway 88? I used to spend a bit of time in that area.

Aug 24, 2017 at 8:57 AM, Nikola Nikolov:

Rob, The location is near Shot Rock Vista campground; close to Silver Lake. I was wrong about the elevation, it is less than 8000 ft, but still above 7000. Horizons are open on all directions.

There is literary nothing around that can cause light pollution. Highway 88 is not visible from the site, so there is very little chance for pollution from headlights. I do not have information about domes. Sacramento is the nearest city, 86 miles (2 hours driving) from the campground; it should have minimal impact. Rogelio Bernal was in the area of Silver Lake in early summer to take night sky photography.

Drawback: the site requires RV, because there is nothing close.

Rob Jaworski Aug 24 9:52 AM

Thanks, Nikola. That looks like it's the site of the old Iron Mountain ski resort that was shuttered many years ago. I seem to recall that the road leading into that area was chained or gated off. Is that not the case? It's also unclear if this is an actual campground as well. Is it marked as such, or is it more of an ask-for-forgiveness kind of deal?

In any event, sites up highway 88 are interesting since there are no close cities or even towns. It can get quite dark there, but you are pretty far out there.



Hubble's Messier Catalog



Charles Messier (1730–1817) was a French astronomer best known for his "Catalog of Nebulae and Star Clusters." An avid comet-hunter, Messier compiled a catalog of deep-sky objects in order to help prevent other comet enthusiasts from wasting their time studying objects that were not comets.

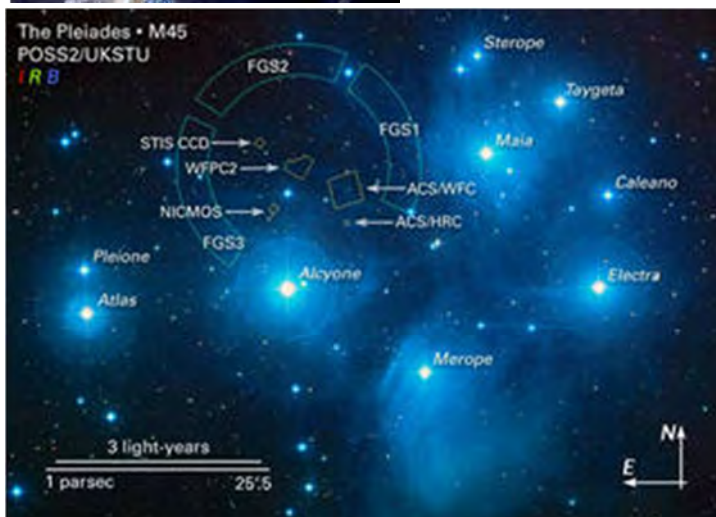
Credits: R. Stoyan et al., *Atlas of the Messier Objects: High-lights of the Deep Sky* (Cambridge University Press, 2008)

Although there are as many as one hundred billion comets in the outer regions of the solar system, prior to 1995, only around 900 had ever been discovered. This is because most comets are too dim to be detected without the proper astronomical equipment. Occasionally, however, a comet will sweep past the sun that is bright enough to be seen during the daytime with the naked eye.

One such instance occurred in 1744. Comet Klinkenberg-Chéseaux, discovered by three amateur astronomers in late 1743, grew steadily brighter as it approached the sun. By the end of February 1744, the comet had reached its peak brightness at an apparent magnitude of -7 , making it the brightest object in the sky except for the sun and moon. The comet's brilliance captured the interest of professional and amateur astronomers alike, including a young Charles Messier.

Born in 1730 in Badonviller, France, Messier had to give up formal education at age 11 when his father died. Soon after, he witnessed the spectacular Comet Klinkenberg-Chéseaux, which ignited his passion for astronomy. At the age of 21, Messier was hired as a draftsman for the French navy. He learned to use astronomical tools and became a skilled observer. For his efforts, Messier was eventually promoted to the chief astronomer of the Marine Observatory in Paris, where he pursued his interest in comets. He discovered over a dozen comets, earning him the nickname "Comet Ferret" from King Louis XV.

In 1758, Messier was in the process of observing one such comet when he was distracted by a cloudy object in the constellation Taurus. Upon further observation, he realized that the object could not be a comet because it was not moving across the sky. In an effort to prevent other astronomers from mistaking the object for a comet, Messier took note of it and began to catalog other comet-like "objects to avoid." (Cont on following page)



The brilliant stars seen in this ground-based image are members of the open star cluster M45, also known as the Pleiades, or Seven Sisters. The shapes overlaid on the image represent the fields of view of Hubble's cameras and other science instruments, and provide a scale to Hubble's very narrow view on the heavens.

Credits: NASA, ESA and AURA/Caltech



The Hubble Space Telescope is equipped to take images in various wavelengths of light in order to provide more insight into its targets. The famous Pillars of Creation in the Eagle Nebula (also known as M16: the 16th object in Charles Messier's catalog) were imaged using both visible (left) and infrared (right) filters. Using infrared light, Hubble is able to probe past the dense gas and dust of the nebula to reveal stars that are hidden in visible wavelengths.

Credits: NASA, ESA/Hubble and the Hubble Heritage Team



This comet-like object that Messier observed was NGC 1952. Commonly known today as M1 (Messier 1) or the Crab Nebula, it is the first object in Messier's Catalog of Nebulae and Star Clusters. By the time of his death in 1817, Messier had compiled a list of 103 objects in the night sky using his own observations with various telescopes and the discoveries of other astronomers. The catalog was revised in the 20th century and now contains 110 objects.

The Messier catalog includes some of the most fascinating astronomical objects that can be observed from Earth's Northern Hemisphere. Among them are deep-sky objects that can be viewed in stunning detail using larger telescopes but are also bright enough to be seen through a small telescope. This characteristic makes Messier objects extremely popular targets for amateur astronomers possessing all levels of experience and equipment. They are so popular, in fact, that they have inspired a special award from the Astronomical League (an organization for amateur astronomers) given to observers who are able to spot each of these objects. Those who succeed receive a certificate and are given the distinction of being in the Messier Club.

While the Hubble Space Telescope has not produced images of every object in the Messier catalog, it has observed 93 of them as of August 2017. Some of Hubble's photographs offer views of a given object in its entirety, but many focus on specific areas of interest. While Hubble is able to magnify objects very effectively, it has a relatively small field of view. This means that, in some cases, Hubble would need to take many exposures to capture an entire object. Although this is not always an efficient use of its time, as is the case for the widely spaced "open" star clusters in the Messier catalog, many exposures are taken when the scientific value justifies the time spent. One of these objects is the Andromeda galaxy (designated M31 in Messier's catalog). In order to create a mosaic image that depicts almost half of Andromeda, Hubble has taken nearly 7,400 exposures of the galaxy.

Unlike a digital camera that takes a single photograph in red, green and blue light to create a single full-color image, Hubble takes monochrome images at specific wavelengths of light. These specific wavelengths can reveal characteristics of an object that are of scientific interest, such as the presence of a particular chemical element. Multiple observations at different wavelengths can be combined to form a single image that reveals all of these characteristics at once but doesn't necessarily contain the full spectrum of visible light. In those cases, colors are assigned to each wavelength to highlight the different characteristics, offering a deeper understanding of the object's properties.

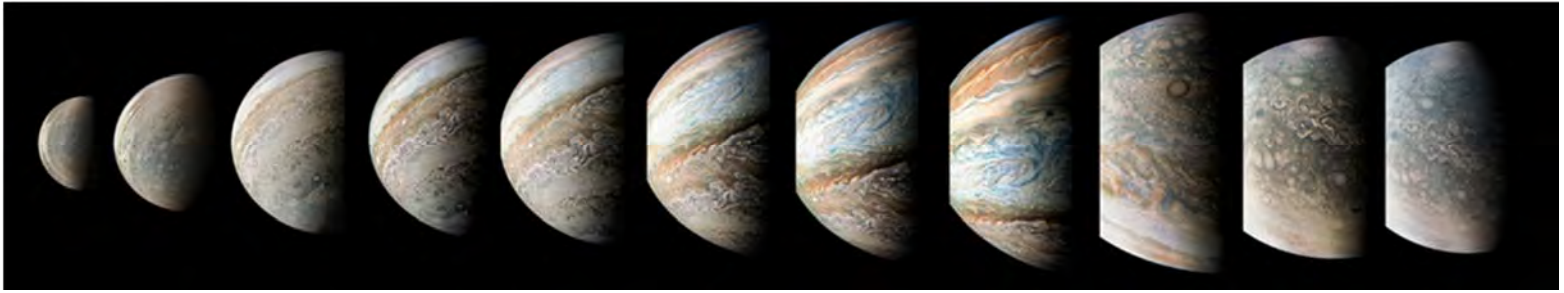
Additionally, Hubble is equipped to take infrared and ultraviolet images, which can reveal information that cannot be obtained using only visible light. Because infrared and ultraviolet light are not visible to human eyes, these images need to be processed in such a way that makes them meaningful to observers. This is done by assigning colors that humans can perceive to the wavelengths that they cannot.

Whether their tool of choice is a sophisticated ground-based telescope, a decent pair of binoculars, or simply their naked eyes, observers hunting for Messier objects can use the data gathered from Hubble's spectacular images to deepen their understanding of these 110 highlights of the night sky as they carry on the tradition of amateur astronomy.

Credit: nasa.gov



95 Minutes Over Jupiter



This sequence of color-enhanced images shows how quickly the viewing geometry changes for NASA's Juno spacecraft as it swoops by Jupiter. The images were obtained by JunoCam. Image Credits: NASA/JPL-Caltech/SwRI/MSSS/Kevin M. Gill

Once every 53 days, Juno swings close to Jupiter, speeding over its clouds. In just two hours, the spacecraft travels from a perch over Jupiter's north pole through its closest approach (perijove), then passes over the south pole on its way back out. This sequence shows 11 color-enhanced images from Perijove 8 (Sept. 1, 2017) with the south pole on the left (11th image in the sequence) and the north pole on the right (first image in the sequence).

The first image on the right shows a half-lit globe of Jupiter, with the north pole approximately at the upper center of the image close to the terminator -- the dividing line between night and day. As the spacecraft gets closer to Jupiter, the horizon moves in and the range of visible latitudes shrinks. The second and third images in this sequence show the north polar region rotating away from the spacecraft's field of view while the first of Jupiter's lighter-colored bands comes into view. The fourth through the eighth images display a blue-colored vortex in the mid-southern latitudes near Points of Interest "[Collision of Colours](#)," "[Sharp Edge](#)," "[Caltech, by Halka](#)," and "[Structure01](#)." The Points of Interest are locations in Jupiter's atmosphere that were identified and named by members of the general public. Additionally, a darker, dynamic band can be seen just south of the vortex. In the ninth and tenth images, the south polar region rotates into view. The final image on the left displays Jupiter's south pole in the center.

From the start of this sequence of images to the end, roughly 1 hour and 35 minutes elapsed. JunoCam's raw images are available for the public to peruse and process into image products at: www.missionjuno.swri.edu/junocam

More Than 2.4 Million Names Are Going to Mars

Last month, NASA invited members of the public to send their names to Mars. And the public responded loud and clear.

More than 1.6 million people signed up to have their names etched on a microchip that will be carried on NASA's upcoming InSight mission, which launches in May of 2018.

NASA's Jet Propulsion Laboratory in Pasadena, California, reopened the opportunity after it proved successful in 2015. During that open call, nearly 827,000 names were collected for a microchip that now sits on top of the robotic InSight lander.

The grand total once a second microchip is added in early 2018 will be 2,429,807 names. Space enthusiasts who signed up this last round shared their downloadable "boarding passes" on social media, complete with the total number of flight miles they've collected by participating in engagement initiatives for other Mars missions.

InSight will be the first mission to look deep beneath the Martian surface, studying the planet's interior by listening for marsquakes. These quakes travel through geologic material at different speeds and give scientists a glimpse of the composition and structure of the planet's inside. The insights into how Mars formed will help us better understand how other rocky planets are created.

Credit: nasa.gov



*Total Solar Eclipse 2017,
Enhanced Corona with Hint of Earthshine*

From: Gary Chock

© 2017 Gary Chock



Location: Oregon Star Party, August 21, 2017 @ 10:21:43

Mount: iOptron ZEQ25GT

Camera: Canon EOS 6D (stock) + EF 100-400mm L IS II + Extender EF 2x III

Exposure: 0.5 seconds at f/11, 800mm, ISO 100

Processing: Adobe Lightroom 5

*I adjusted for corona detail and a hint of Earthshine. The blue dot below is Regulus (Alpha Leonis). I was using exposure bracketing and an intervalometer, taking a set of shots every 3 seconds from 2nd Contact to 3rd Contact.
Gary*





Subjects: *(clockwise)* Elephant Trunk Nebula, Fish Head Nebula, Pacman Nebula

By: Everett Quebral

Location: Northern Calif.

Date: September 24th and 25th, 2017

Telescope & Guide scope: Orion 80mm ShortTube with Orion SSAG for guiding.

Filters: Orion 7nm bandpass. Orion ED80T-CF with a ZWO ASI1600MM-Cool, ZWO 8 position filter wheel.

Images: 12 sub x 5 minutes exposures per channel. 3 hours of total integration.

Programs: Sequence Generator Pro, PixInsight 1.8 and Photoshop CC2017.

Remarks: The post processing that I have done is very quick, just PixInsight pre processing with 30 bias frames, 12 dark frames and no flats. LRGB combination with ColorMask and Curves to remove star halo. Finally Photoshop for last minor noise and hue/saturation correction.

I used the Hubble Palette, S2HaO3.



Elephant's Trunk Nebula

By Alan Pham

Locations: San Jose; Pinnacles National Park, California

Dates: May 20, 2017, May 27, 2017

Telescope: Stellarvue SV4
Imaging camera: ZWO ASI1600MM-Cool
Mount: Orion Sirius EQ-G
Guide scope: Orion Mini 50mm
Guiding camera: QHY5L-II-M
Focal reducer: Astro Tech 0.8x Reducer/Flattener

Filters / Frames:
Astrodon Filters B, G, R Tru-Balance E-Series Gen 2: 44x60" bin 1x1
Astrodon Ha 1.25 5nm: 200x180" bin 1x1
Integration: 12.6 hours
Software: PixInsight 1.8, PHD2
Remarks: This image was processed with techniques to make it look more like a pure RGB image.

Copyright Alan Pham



The Tadpoles

From Glenn Newell

Location: Scotts Valley, Ca.

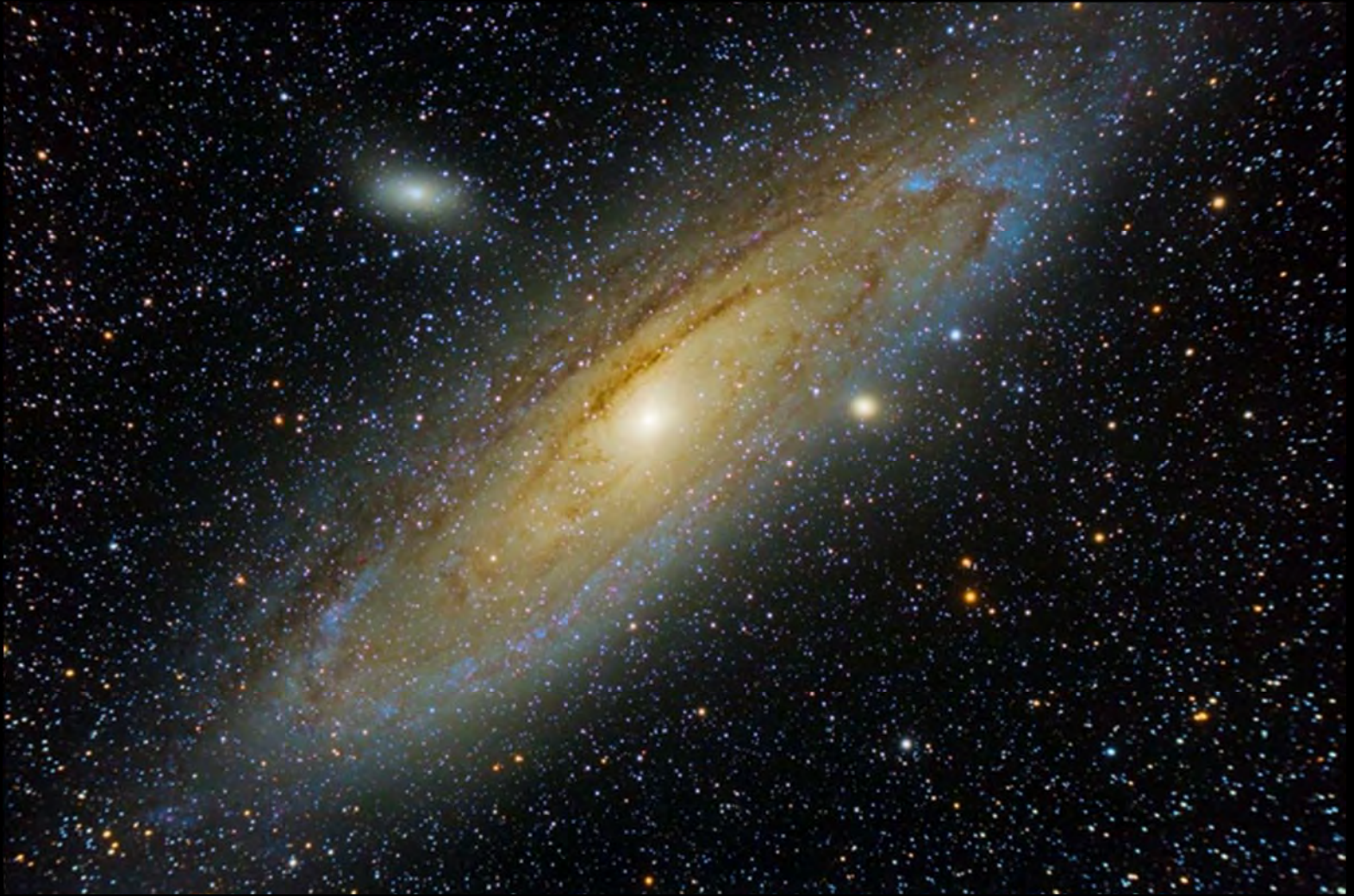
Date: Oct. 3, 2017

Telescope: TPO 12" Truss RC
Mount: Orion HDX110 EQ-G
Imaging camera: ZWO ASI1600MM-Cool
Guiding camera: QHYCCD QHY5L-II mono
Focal reducer: Astro-Physics CCDT67
Software: Adobe Photoshop CS6, Topaz Labs' DeNoise, CCDStack2
Filters: Oiii, Blue, Green, Red, Ha, Sii
Resolution: 2328x1604
Frames: Blue: 7x300" (gain: 300.00) -26C bin 2x2
Ha: 97x300" (gain: 300.00) -26C bin 2x2
Red: 7x120" (gain: 300.00) -26C bin 2x2
Integration: 15.7 hours, Darks: ~25

Green: 7x300" (gain: 300.00) -26C bin 2x2
Oiii: 48x300" (gain: 300.00) -26C bin 2x2
Sii: 26x300" (gain: 300.00) -26C bin 2x2



SJAA Member Astrophoto Gallery



M31 Andromeda Galaxy

By: Ahmed Sakr

Location: Mendocino, Ca

Date: July 23, 2017

Comment: This is my first real image and I am really proud of it.

Optics: SV80ST

Camera: T3i/ 280 seconds @ 400 ISO

Guiding: used SSAG from Orion.

42 lights, 20 darks

Processing: Pixinsight.





Kid Spot Jokes:

- ♦ **Why did the star get arrested?**
Because it was a shooting star!
- ♦ **What happened to the astronaut who stepped on chewing gum ?**
He got stuck in orbit!
- ♦ **How does one astronaut on the moon tell another astronaut that he is sorry ?**
He Apollo-gises!

Kid Spot Quiz:

1. Catadioptric telescopes use?
2. What is the f-number of a telescope?
3. How is the light gathering power of an optical telescope is related to the aperture?

100 Years of Astronomical Discovery

Excerpts From Astronomy Magazine September 2016 Edition

1930 Clyde Tombaugh discovers Pluto

1932 Estonian astronomer Ernst Opik proposes that long-period comets originate in a distant region of the solar system now known as the Oort Cloud, named after Jan Oort, who will independently suggest the idea in 1950.

1933 The birth of radio astronomy. Karl Jansky (Bell Labs), charged with figuring out what was causing interferences with short wave communications, builds a directional antenna and soon discovers that the interference is coming from far outside the solar system. After correlating his observations with astronomical charts he pinpointed the mystery signals as coming from the constellation Sagittarius, concluding that the signals were coming from the center of the Milky Way Galaxy.

1933 Swiss astronomer Fritz Zwicky proposes dark matter as the gravitational glue that keeps galaxies in the vast Coma Cluster bound together.

1935 Public Astronomy Comes of Age. Both Griffith Observatory in Los Angeles and New York's Hayden Planetarium open in 1935. The Griffith Observatory facility included a Zeiss refracting telescope with a 12" objective lens.

1936 The mirror that would become the glass giant of Palomar and open a universe of wonder leaves the Corning Glass Works in Corning, New York, on March 26 for a 16 day train ride across America.

1938 War of the Worlds.....With a script based loosely on the H.G. Wells' novel the *The War of the Worlds*, a young Orson Welles planned to give the American People a bit of a scare with the 1938 broadcast. For many people (who missed the opening) widespread panic ensued for fear of an invasion from Mars.

Kid Spot Quiz Answers:

1. Both lenses and mirrors
2. Ratio between focal length and aperture of primary mirror or lens
3. Square of the aperture of primary mirror or lens

From the Board of Directors

Announcements

- ◇ Volunteers for the School Star Party program are needed!
- ◇ SJAA Club ByLaws have been changed. Please see Rob Jaworski article on the following page.
- ◇ Annual Show-n-Tell event is scheduled for Saturday, December 2, 2017 after the Board Meeting.
- ◇ The 2018 Royal Astronomical Society of Canada (RASC) handbooks and calendars will be for sale starting December 2nd after the board meeting. Handbooks \$20 and calendars \$15 each.

Board Meeting Excerpts

September 9, 2017

In attendance

Teruo Utsumi, Bill O'Neil, Rob Jaworski, Rob Chapman, Sukhada Palav, Vini Carter, Glenn Newell, Swami Nigam
Excused: Wolf Witt
Guest: None

Hogue Park Application to San Jose

The City of San Jose has accepted the SJAA RFQ. Rob Jaworski is searching for a CPA to handle the audit portion of the application. Teruo Utsumi will keep the board updated on any additional requirements.

Calstar Insurance Rider

Teruo Utsumi noted that the SJAA insurance rider for Calstar will be changed from Mt. Hermon to Whisper Canyon to accommodate the new event location. SJAA is a partner in the event and they make use of the SJAA insurance.

Mendoza Ranch Permit

The Mendoza Ranch permit is up for renewal. The \$300 permit fee expense was unanimously approved by the board. Ed Wong will handle the re-application to the County.

SJAA 2018 Calendar

The SJAA 2018 calendar was motioned and approved by the board.

Open Space Authority (OSA) Requirements

OSA has revisited their rule change request for RCDO star parties and decided that most of the proposed rules would not be required. OSA/SJAA plan to start up Starry Nights again with a reservation system and cap for the number of people allowed to attend. In exchange for SJAA astronomers at the Starry Nights, so long as an OSA document present, SJAA will be allowed to use OSA land for club use with the required notice. Sukhada Palav will follow up on finding a new SJAA/OSA liaison to replace Dave Ittner / Ed Wong who are temporarily filling this role.

Complimentary / Lifetime Memberships

Rob Jaworski proposed that the board consider providing free memberships to active members which would be approved by the board on a case by case basis. No decision was made. A formal set of requirements needs to be developed.

October 7, 2017

In attendance

Bill O'Neil, Swami Nigam, Rob Chapman, Sukhada Palav, Glenn Newell, Excused: Teruo Utsumi, Vini Carter, Wolf Witt, Rob Jaworski
Guest: Maya

New Production Editor - Ephemeris

Per Bill O'Neil, Ephemeris Production Editor Tom Piller is retiring after 4 years of creating an excellent SJAA chronicle. There are no candidates who have showed immediate interest. Discussion ensued on how to fill the vacancy and the future format of the publication.

November, 2017

In attendance

Bill O'Neil, Rob Jaworski, Wolf Witt, Rob Chapman, Glenn Newell, Vini Carter, Swami Nigam, Sukhada Palav, Absent Excused: Teruo Utsumi, Guest: None

Hogue Park Application to San Jose

Teruo Utsumi will report CoSJ contract status at a later date; the City has accepted SJAA's RFQ. Rob Jaworski is searching for a CPA to handle the audit portion of the application. Teruo Utsumi will keep the board

updated on any additional requirements.

New Member Personal Info

Membership Chair Dave Ittner requested we discuss what information we request from New Members; discussion followed but no consensus was reached on how to proceed.

Discussion summarized as follows.... SJAA will continue to collect member home address for the loaner program and on the membership signup form along with email addresses for communicating with members. The proposal to make the home address request an optional field on the online signup did not have enough votes to pass.

Swap Meet Report

Rob Jaworski / Vini Carter chaired the event. Rob Jaworski managed the financial aspects and Vini Carter managed operations. Much of the board attended.

Board discussion followed on seller setup start time versus when the buyers are allowed to enter. Vini's preference was to allow sellers come in first and get setup before we open the door to buyers. Prior recommendations were that SJAA open the door to sellers and buyers at the same time. The board was generally okay with either, so we will leave it up to the event chair to decide.

No immediate volunteers for spring swap meet. We are deferring a decision on hosting a spring auction until spring is closer.

JVN Thank You Scope

Rob Jaworski clarified the details of a telescope JVN gift versus school program acquisition and requested additional funding for the purchase. Per Rob, there is a need to clarify whether the purchase is a personal thank-you gift to JVN (as we agreed was the original intent) or a telescope for the school program (which is what the original June 2017 agenda item made it sound like). Rob J. proposed that it be given to JVN for as long as he wants it. After JVN no longer has need for the scope, ownership would revert to the SJAA school star party program. This idea was approved by the board along with the additional funding required to complete the equipment purchases.

Amending Bylaws: Memberships

Posted on August 31, 2017 by Rob Jaworski



At the board of directors meeting, July 2017, the board voted to amend the bylaws. This kind of move is not done very often, and great consideration is made before such an idea is even brought to a vote by the board. However, given the history of the types of memberships with the SJAA, and also some recent moves made by the City of San Jose, the board deemed it prudent to amend the bylaws to remove the option of offering youth or student memberships, and to stipulate that only people age 18 and above are able to apply for membership.

Specifically, the board voted to remove Article 12, Section 1, Part C in its entirety. Further, Article 12, Section 2 was amended to indicate only people 18 and over can be members.

Over the past seven or eight years during which I have been on the board of the SJAA, I have served as membership chair at least twice. Also, at every board meeting each month, the membership report lists all those individuals who have applied for new or renewing membership. Plainly, there is a good amount of visibility of the member rolls. During those years, I have seen not more than a small handful of members who applied via the student/youth membership. The number has been, at

most, in the single digits.

Recently, we have seen the City of San Jose put forth some requirements which could be quite burdensome to SJAA volunteers. In the recent requalification process that the City required of its reuse partners (of which, SJAA is one), it called for all volunteers or employees that has direct contact with minors to have background checks run, TB tests completed, fingerprints taken, and all the other things that typically come with interacting with children. On the face of it, everyone in the club that shows up to In Town Star Parties or Solar Sundays, or any of the myriad events that the SJAA hosts at Houge Park, could be considered a volunteer that has potential interaction with children. It would be too much to ask of SJAA volunteers to agree to such a requirement, and it could potentially put the events at Houge Park at risk.

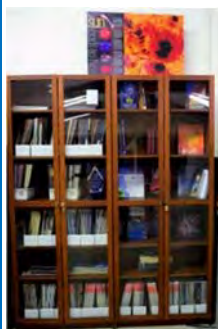
Upon further research and investigation, the board concluded that although the contract with the City does not provide a definition of *direct contact*, generally, volunteers of SJAA do not have *direct contact*. Rather, SJAA volunteers have *incidental contact* with minors. Kids may show up to SJAA events with their parents or guardians, but SJAA volunteers never take custody of or supervise them, such as in an afterschool or day care environment. The conclusion was made that since no one that is associated with SJAA ever has direct contact with minors, the organization is exempt from such requirements.

To further solidify the fact that SJAA does not provide care, supervision, guidance or control of minors in any way, the board thought it prudent that minors be prohibited from becoming members in the first place.

The board does not believe that these changes to the bylaws will have any impact whatsoever to its members, now or in the future. SJAA will recognize the benefit, however, of not being potentially held to onerous requirements of its volunteers, nor will there be any question about the club's relationship to minors. The club certainly encourages kids of all ages to learn more about astronomy and the sciences and to come to club events, and SJAA will continue to support kids's desires to learn more, or to just look at Saturn through a telescope. But please, do bring a parent or guardian!



SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held every other month.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a

similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops & Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about. Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.



<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Teruo Utsumi
Vice Pres/Dir:	Bill O'Neil
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Rob Chapman
Director:	Vini Carter
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Swami Nigam
Director:	Sukhada Palav
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Tom Piller
Fix-it Program:	Vini Carter
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Vinod Venkateshan
Loaner Program:	Muditha Kanchana
Memberships:	Dave Ittner
Publicity:	Rob Jaworski
Questions:	The Board
Quick START	Dave Ittner
Solar & Starry Nights:	Dave Ittner
School Events:	Jim Van Nuland
Speakers:	Sukhada Palav
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by not later than the 20th of the month of February, May, August and November. (earlier is better).

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

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San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

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San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

- ☐ **New** ☐ **Renewal** (Name only if no corrections)
- ☐ **\$20** Regular Membership with **online** Ephemeris
- ☐ **\$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

Name: _____

Address: _____

City/ST/Zip: _____

Phone: _____

E-mail address: _____

